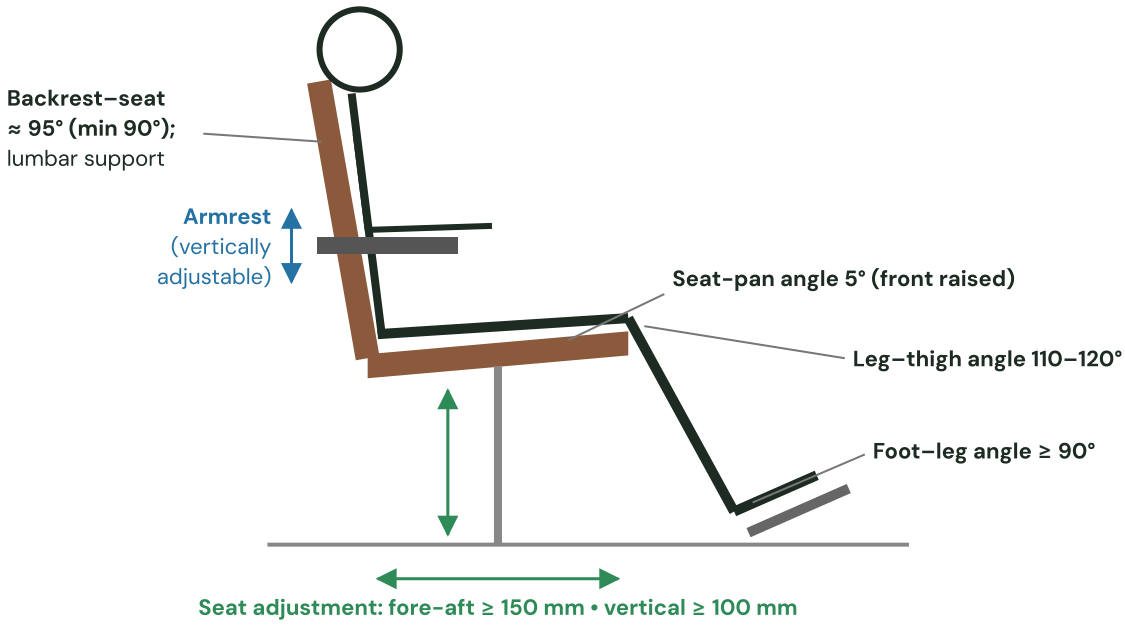
UNIT 5 · TOPIC 10(A)

Seat Design and Seat Suspension

Syllabus topic: Visual range and limitations; seat design

Dimensional prescriptions (Liljedahl pp. 226–227)

SAE J899 seat dimensions • J1163 seat location • J898 comfort & reach zones (Liljedahl pp. 226–227)



Operator seat geometry (redrawn from the source figure)

Feature	Prescription
Seat-cushion (pan) angle	**5°** from horizontal, front of cushion raised — supports thighs and upper body
Backrest–seat angle	≈ **95°**, minimum 90°
Backrest	MUST support the LUMBAR back
Armrests	Needed for shoulder support; VERTICALLY ADJUSTABLE for forearm/elbow freedom at hand controls
Leg-to-thigh angle (pedal operation)	**110–120°**
Foot-to-leg angle	≥ **90°**
Seat adjustment	≥ **150 mm** fore-and-aft; ≥ **100 mm** vertical

Governing standards (p. 227)

Standard	Covers
SAE J899	Operator seat dimensions
SAE J1163	Seat location relative to platform and controls
SAE J898	Zones of comfort and reach for control location
SAE J185	Steps, handrails and handholds for platform access
SAE J154a	Minimum operating "space envelope" around the operator in a cab

- **Why it matters:** WBV plus a cramped posture produced the 71.3% spine-deformation figure among tractor drivers (Topic 6b, p. 212).
- The cushion should spread pressure over the ischial tuberosities, allow posture change, and use breathable material for Indian heat.

Seat-suspension guidelines (pp. 223–224)

Passive suspension uses constant- or variable-rate springs, shock absorbers for low-frequency damping, and a torsion bar or pneumatic cylinder to compensate static deflection (operator weight).

Parameter	Design target
Undamped natural frequency of the seat suspension	< **1.5 Hz**
Seat f_n relative to tyre f_n	**0.4–0.5 ×**
Maximum suspension travel	≈ **76 mm**
Isolation performance (transmissibility)	**0.4–0.5** (Topic 6b gives the satisfactory range 0.4–0.65)
Best ride from tyres	Use the LARGEST-diameter tyres — they negotiate obstacles more gently and act as a filter (p. 224)

The central compromise (p. 224): a softer suspension isolates better, but lets the operator move further relative to the controls.

- Fore-aft isolators on linear bearings, and roll/pitch isolators, address the other axes (Rakheja & Sankar).

Practice MCQs

Q106. The recommended seat-cushion (pan) angle from horizontal is:

(a) 0° (b) 5° with front raised
(c) 15° (d) 25° rear raised

Q107. The angle between backrest and seat should be approximately:

(a) 75° (b) 90° exactly
(c) 95° (minimum 90°) (d) 120°

Q108. The backrest must specifically support the:

(a) Neck only (b) Lumbar back
(c) Shoulders only (d) Head

Q109. For pedal operation the leg-to-thigh angle should be:

(a) 60–80° (b) 90–100°

(c) 110–120° (d) 150–170°

Q110. The minimum foot-to-leg angle is:

(a) 45° (b) 60°

(c) 90° (d) 120°

Q111. Minimum fore-and-aft seat adjustment is:

(a) 50 mm (b) 100 mm

(c) 150 mm (d) 300 mm

Q112. Minimum vertical seat adjustment is:

(a) 25 mm (b) 100 mm

(c) 200 mm (d) 400 mm

Q113. Armrests must be:

(a) Fixed (b) Vertically adjustable for varying operator sizes

(c) Removed (d) Padded only

Q114. Operator's seat DIMENSIONS are given in:

(a) SAE J899 (b) SAE J209

(c) ASAE S335.3 (d) ISO 500

Q115. Zones of comfort and reach for control location are defined by:

(a) SAE J898 (b) SAE J185

(c) IS 4468 (d) ASAE S383.1

Q116. Dimensions for seat location relative to platform and controls:

(a) SAE J1163 (b) SAE J154a

(c) ISO 730 (d) IS 9020

Q117. Steps, handrails and handholds are covered by:

(a) SAE J185 (b) SAE J899

(c) ASAE EP443 (d) ISO 2631

Q118. The minimum operating 'space envelope' around the operator in a cab is guided by:

(a) SAE J154a (b) SAE J209

(c) IS 6284 (d) ASAE S304.5

Q119. Undamped natural frequency of a seat suspension should be:

(a) <1.5 Hz (b) 3–5 Hz

(c) 8–10 Hz (d) 20 Hz

Q120. Seat natural frequency should be what fraction of tyre natural frequency?

(a) 0.1 (b) 0.4–0.5

(c) 1.0 (d) 2.0

Q121. Maximum seat suspension travel is about:

(a) 25 mm (b) 76 mm

(c) 150 mm (d) 300 mm

Q122. Target seat isolation performance (transmissibility) is:

(a) 0.05 (b) 0.4–0.5

(c) 1.0 (d) 2.0

Q123. The penalty of a very soft seat suspension is:

(a) Poor isolation
(b) Larger operator displacement relative to the controls
(c) Higher cost only
(d) Noise

Q124. Static deflection due to operator weight is compensated by:

(a) Torsion bar or pneumatic cylinder
(b) Extra padding
(c) Steel plate
(d) Nothing

Q125. For best ride comfort, tyres should be:

(a) Smallest diameter
(b) Largest diameter
(c) Solid
(d) Narrow

Answers

Q106 — (b) 5° with front raised — *Liljedahl pp. 226–227 — memorise 5°, 95°/90°, 110–120°, ≥ 90°, 150 mm, 100 mm.*

Q107 — (c) 95° (minimum 90°)

Q108 — (b) Lumbar back

Q109 — (c) 110–120°

Q110 — (c) 90°

Q111 — (c) 150 mm

Q112 — (b) 100 mm

Q113 — (b) Vertically adjustable for varying operator sizes

Q114 — (a) SAE J899 — *p. 227 — SAE family: J899 dimensions, J1163 location, J898 comfort/reach zones, J185 access, J154a space envelope.*

Q115 — (a) SAE J898

Q116 — (a) SAE J1163

Q117 — (a) SAE J185

Q118 — (a) SAE J154a

Q119 — (a) <1.5 Hz

Q120 — (b) 0.4–0.5

Q121 — (b) 76 mm

Q122 — (b) 0.4–0.5

Q123 — (b) Larger operator displacement relative to the controls — *p. 224 — the softness/displacement compromise.*

Q124 — (a) Torsion bar or pneumatic cylinder

Q125 — (b) Largest diameter — *p. 224 — the largest tyres filter terrain best.*

Sources: Liljedahl, Turnquist, Smith & Hoki — Tractors and their Power Units, Ch. 9 (spatial/visual/control requirements pp. 226–231; seating pp. 219–224); SAE J899 / J1163 / J898 / J185 / J154a / J209; ASAE S335.3, S304.5 / SAE J389b, EP443; Purcell (1980), Stikeleather (1981); NPTEL lectures.